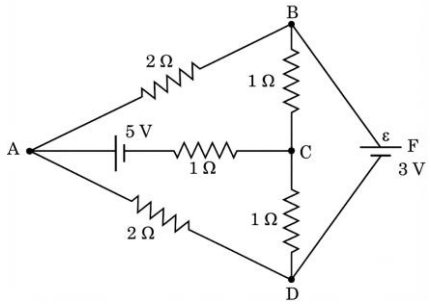
B.A.T. (Baseline Assessment Test)

No. of Question : 50 **Date : 26 Sep. 2026** **Time Duration : 75 Min.**

Topic : CURRENT ELECTRICITY

1. The magnitude of the electric current flowing through the branch CD is :



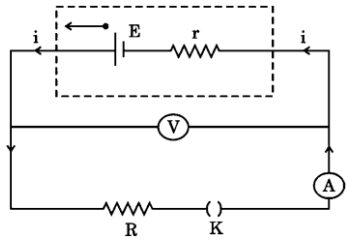
(1) 0.5 A

(2) 0 A

(3) 3 A

(4) 4.5 A

2. A battery of unknown emf E and internal resistance r is connected in a circuit. When the key (K) is open, the voltmeter reads 10.0 V and the ammeter reads 0 A. In the closed circuit, the voltmeter reads 6.0 V and the ammeter reads 2.0 A. The values of internal resistance (r) and external resistance (R) are respectively:



(1) 2.0 Ω, 3.0 Ω

(2) 3.0 Ω, 2.0 Ω

(3) 4.0 Ω, 1.5 Ω

(4) 1.0 Ω, 5.0 Ω

3. **Assertion (A) :** When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.

Reason (R) : The drift velocity is superposed over large random velocities of electrons.

(1) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(2) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(3) If Assertion (A) is true and Reason (R) is false.

(4) If both Assertion (A) and Reason (R) are false.

4. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is :

(1) 2 Ω (2) 1.2 Ω

(3) 1 Ω (4) 0.5 Ω

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5. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

(1) P (2) $\frac{P}{2}$

(3) $\frac{P}{4}$ (4) 2P

6. A galvanometer of resistance 50 Ω is converted into a voltmeter of range (0–2 V) using a resistor of 1.0 kΩ. If it is to be converted into a voltmeter of range (0–10 V), the resistance required will be :

(1) 4.8 kΩ (2) 5.0 kΩ

(3) 5.2 kΩ (4) 5.4 kΩ

Space for Rough Work

7. Two wires P and Q are made of the same material. The wire Q has twice the diameter and half the length as that of wire P. If the resistance of wire P is R, the resistance of the wire Q will be

(1) R (2) $\frac{R}{2}$

(3) $\frac{R}{8}$ (4) 2R

8. A conductor of length l is connected across an ideal cell of emf E. Keeping the cell connected, the length of the conductor is increased to 2l by gradually stretching it. If R and R' are initial and final values of resistance and v_d and v_d' are initial and final values of drift velocity, find the relation between R' and R, and v_d' and v_d.

(1) $R' = 2R$ and $v_{d'} = \frac{v_d}{4}$

(2) $R' = 4R$ and $v_{d'} = \frac{v_d}{2}$

(3) $R' = 4R$ and $v_{d'} = 2v_d$

(4) $R' = \frac{R}{4}$ and $v_{d'} = \frac{v_d}{2}$

9. n identical cells, each of e.m.f. E and internal resistance r, are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'.

(1) $E\left(1-\frac{4}{n}\right)$

(2) $2E\left(1-\frac{2}{n}\right)$

(3) $2E\left(1-\frac{4}{n}\right)$

(4) $E\left(1+\frac{2}{n}\right)$

10. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is

(1) 1.05 (2) 2

(3) 0.50 (4) 1.5

11. When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :

(1) the electrons coming from the power source move fast through the initially empty filament.

(2) the filament may be old having low resistance.

(3) electric field is established instantaneously across the filament which pushes the electrons.

(4) free electrons in the filament travel with the speed of light.

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12. A voltmeter of resistance 1000 Ω can measure up to 25 V. How will you convert it so that it can read up to 250 V ?

(1) By connecting a resistance of 9000 Ω in parallel

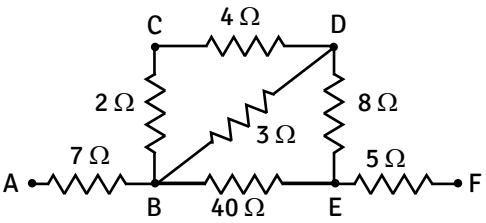
(2) By connecting a resistance of 9000 Ω in series

(3) By connecting a resistance of 10000 Ω in series

(4) By connecting a resistance of 1000 Ω in parallel

Space for Rough Work

13. Find the effective resistance of the network of resistors between points A and F as shown in the figure.



(1) 16 Ω (2) 20 Ω

(3) 24 Ω (4) 12 Ω

14. A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

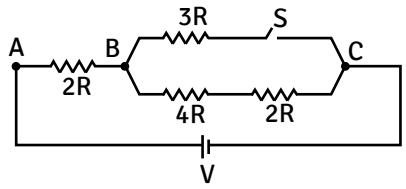
(1) $\frac{P}{4}$

(2) $\frac{P}{2}$

(3) P

(4) 2P

15. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



(1) $\frac{1}{2}$

(2) 2

(3) $\frac{1}{4}$

(4) 1

16. Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

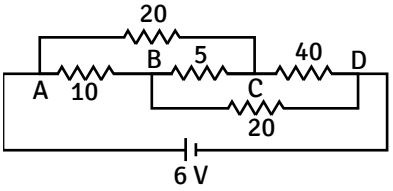
(1) I decreases and II increases.

(2) I increases and II is almost constant.

(3) Both I and II increase.

(4) I decreases and II is almost constant.

17. Calculate the value of the current passing through the battery in the given circuit diagram.



(1) 0.6 A

(2) 0.2 A

(3) 0.3 A

(4) 1.2 A

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18. A copper wire of cross-sectional area $A = 1.0 \times 10^{-7}$ m^2 carries a steady current of I = 2.0 A. Assuming each copper atom contributes one conduction electron and the number density of free electrons is $n = 1.0 \times 10^{29}$ m^{-3} (with electronic charge $e = 1.6 \times 10^{-19}$ C), what is the average drift speed of the conduction electrons?

(1) 2.50 mm s^{-1}

(2) 1.25 mm s^{-1}

(3) 0.625 mm s^{-1}

(4) 5.00 mm s^{-1}

Space for Rough Work

19. A cubical network consists of 12 identical resistors, each of resistance R = 6 Ω, connected across its body diagonal. What is the equivalent resistance of the network between those opposite corners?

(1) 7.5 Ω (2) 3 Ω

(3) 6 Ω (4) 5 Ω

20. A storage battery of a car has an emf of 12 V and an internal resistance of 0.5 Ω. What is the maximum current that can be drawn from this battery?

(1) 12 A

(2) 24 A

(3) 6 A

(4) 18 A

21. A battery of emf 20 V and internal resistance 2Ω is connected to an external resistor R. If the current in the circuit is 2.0 A, what is the value of the resistance R and the terminal voltage of the battery, respectively?

(1) 10 Ω and 18 V

(2) 8Ω and 20 V

(3) 8Ω and 16 V

(4) 6Ω and 12 V

22. At room temperature (27 °C), the resistance of a heating element is 50 Ω. What is the temperature of the element if its resistance increases to 60 Ω, given that the temperature coefficient of the material is $\alpha = 5.0 \times 10^{-3}\ °C^{-1}$?

(1) 57 °C

(2) 67 °C

(3) 77 °C

(4) 87 °C

23. A negligibly small current is passed through a uniform wire of length l = 10 m and cross-sectional area $A = 2.0 \times 10^{-7}\ m^2$. If its measured resistance is R = 5.0 Ω, what is the resistivity of the material at the temperature of the experiment?

(1) $2.0\times10^{-7}\ \Omega\cdot m$

(2) $5.0\times10^{-7}\ \Omega\cdot m$

(3) $1.0\times10^{-7}\ \Omega\cdot m$

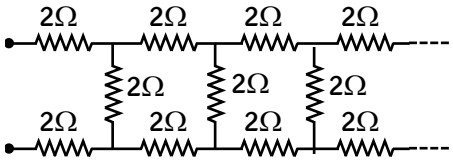
(4) $1.0\times10^{-6}\ \Omega\cdot m$

24. A set of 'n' equal resistors, of value 1Ω each, are connected in series to a battery of emf E and internal resistance 1Ω. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10 I. The value of 'n' is:

(1) 11 (2) 9

(3) 10 (4) 20

25. The equivalent resistance of the infinite network given below is:



(1) $(1+\sqrt{3})\ \Omega$ (2) $(2+\sqrt{3})\ \Omega$

(3) $2(1+\sqrt{3})\ \Omega$ (4) $4(1+\sqrt{3})\ \Omega$

26. Two resistors of resistance, 120 Ω and 300 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 120 Ω to that in 300 Ω in a given time is:

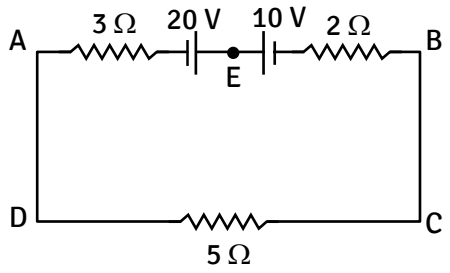
(1) 2 : 5 (2) 5 : 2

(3) 2 : 3 (4) 3 : 2

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Space for Rough Work

27. The magnitude and direction of the current in the following circuit is:



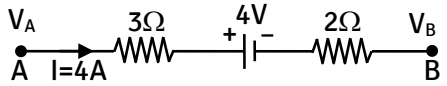
(1) 1 A from A to B through E

(2) 1 A from B to A through E

(3) $\frac{2}{3}$ A from A to B through E

(4) $\frac{5}{7}$ A from B to A through E

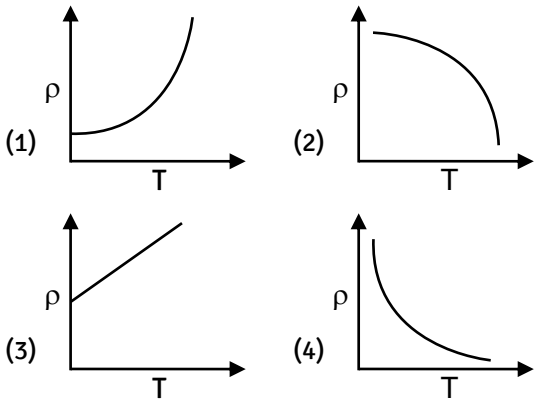
28. The potential difference ($V_A - V_B$) between the points A and B in the given figure is:



(1) +16 V (2) −24 V

(3) +24 V (4) +20 V

29. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?



30. A coil takes 10 min to boil a certain amount of water, another coil takes 20 min for the same process. Time taken to boil the same amount of water when both coil are connected in series

(1) 30 min (2) 6.67 min

(3) 10 min (4) 15 min

31. Consider the following two statements:

(A) Kirchhoff's junction law follows from conservation of charge.

(B) Kirchhoff's loop law follows from conservative nature of electric field.

(1) Both A and B are correct.

(2) A is correct but B is wrong.

(3) B is correct but A is wrong.

(4) Both A and B are wrong.

32. The current through a wire depends on time as $i = i_0 + \alpha t$,

where i_0 = 10 A and α = 4 A s^{-1}. Find the charge crossed through a section of the wire in 10 seconds.

(1) 100 C

(2) 200 C

(3) 250 C

(4) 300 C

33. A charged particle is having drift velocity of 7.5×10^{-4} m s^{-1} in an electric field of 3×10^{-9} V m^{-1}. The electron mobility is

(1) 2.5×10^{4} m^2 V^{-1} s^{-1}

(2) 2.5×10^{5} m^2 V^{-1} s^{-1}

(3) 2.25×10^{-13} m^2 V^{-1} s^{-1}

(4) 4.1×10^{3} m^2 V^{-1} s^{-1}

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Space for Rough Work

34. Which among the following statements is correct?

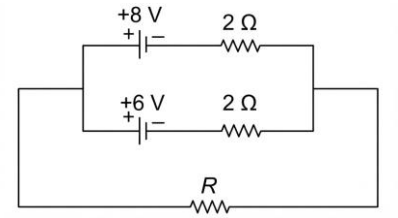
(1) In a metal, number density is independent of temperature

(2) With increase in temperature, relaxation time in metal decreases

(3) For semiconductors and insulators number density increases with increase in temperature

(4) All the above

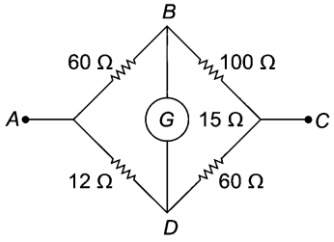
35. In the combination of two cells in parallel by joining positive terminals together and similarly two negative ones, the value of $\frac{E_{eq}}{r_{eq}}$ in circuit is



(1) 7 A (2) 10 A

(3) 2 A (4) 8 A

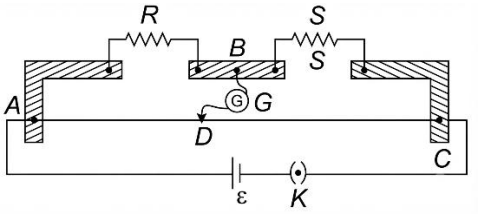
36. Four arms of wheat-stone bridge have the following resistances, AB = 60 Ω, BC = 100 Ω, CD = 60 Ω, DA = 12 Ω. A galvanometer of 15 Ω is connected across BD. Calculate the value of additional resistance connected across CD to balance the bridge.



(1) 12 Ω (2) 15 Ω

(3) 18 Ω (4) 30 Ω

37. In a Meter Bridge null point is found to be at 30 cm from end A. If now a resistance of 10 Ω is connected in parallel with S, the null point occurs at 65 cm, value of S is nearly



(1) 20 Ω

(2) 28 Ω

(3) 33 Ω

(4) 38 Ω

38. When a steady current flows through a conductor of non-uniform cross-section. A physical quantity which does not remain constant is

(1) Drift Velocity

(2) Current density

(3) Electric field

(4) All of these

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39. Three copper wires have length and masses as (l, m), (2l, m/2) and (l/2, 2m). The resistance is minimum in

(1) wire of mass 2m

(2) wire of mass m/2

(3) wire of mass m

(4) same in all three wires

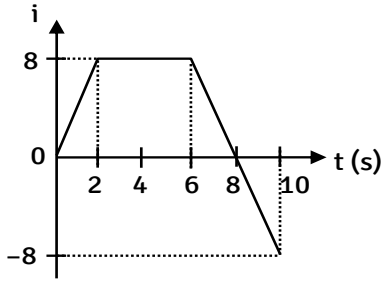
40. Three cells of emfs E, 2E and 3E with equal internal resistance are connected in parallel combination. The equivalent emf of the combination is

(1) E/2 (2) 2E

(3) 6E (4) 6E/11

Space for Rough Work

41. The current i through a conductor varies with time t as shown in figure. The electric charge in interval t = 2 to t = 10 s will be



(1) 32 C

(2) 40 C

(3) 24 C

(4) 16 C

42. If a copper wire is stretched to increase its length by 30%. The percentage increase in resistance of the wire is :

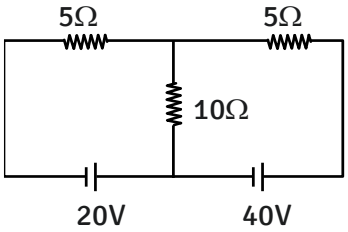
(1) 30%

(2) 60%

(3) 69%

(4) 75%

43. Find current through 10 Ω resistor.



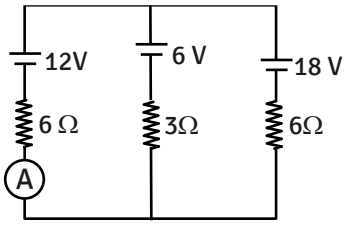
(1) 0.8 A

(2) 2.4 A

(3) 1.6 A

(4) 0.4 A

44. Find the reading of Ammeter in the given circuit.



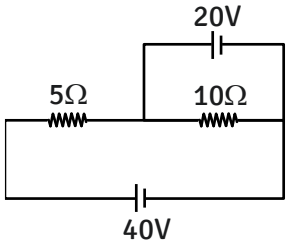
(1) 0.5 A

(2) 0.25 A

(3) 1 A

(4) 2 A

45. Find current in 5Ω resistor



(1) 4 A

(2) 2 A

(3) 8 A

(4) 0 A

46. Three electric bulbs 25 W, 50 W and 100 W are designed to work on a 220 V mains. Which bulb will glow most brightly if they are connected in series across 220 V mains:

(1) 100 W bulb

(2) 50 W bulb

(3) 25 W bulb

(4) all bulbs will glow equally brightly

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Space for Rough Work

47. A 40 W, 200 V, a 60 W, 240 V, and a 100 W, 250 V bulb are connected in parallel across a 400 V supply. What will happen?

(1) Only 40 W bulb will fuse

(2) Only 60 W bulb will fuse

(3) All three bulbs will fuse

(4) None of the bulbs will fuse

48. If a metal wire is stretched to increase its length by 2%. The percentage increase in resistance of the wire is :

(1) 1%

(2) 2%

(3) 3%

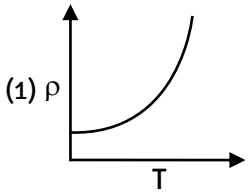
(4) 4%

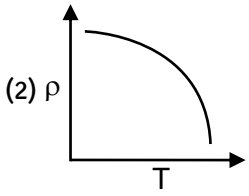
49. The charge Q passing through a cross-section of a conductor varies with time t according to the equation $Q = 3t^2 - 4t + 5$ (where Q is in coulombs and t is in seconds). The instantaneous current at t = 3s is :

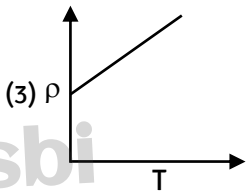
(1) 10 A (2) 12 A

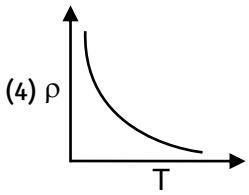
(3) 14 A (4) 16 A

50. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for semiconductor ?

(1) 

(2) 

(3) 

(4) 

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Space for Rough Work